大学物理甲 2 习题解答

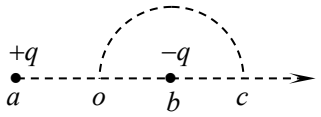
9.24 （1） $U_o=\dfrac{1}{4\pi\varepsilon_0}(\dfrac{q_1}{r}+\dfrac{q_2}{r}+\dfrac{q_3}{r}+\dfrac{q_4}{r})=9\times10^9\,\dfrac{4\times2.0\times10^{-9}}{5.0\times10^{-2}}=1.44\times10^3\,(\mathrm{V})$

（2） $A=q_0(U_\infty-U_o)=-q_0U_o=-1.44\times10^{-6}\,(\mathrm{J})$

（3） $\Delta W=W_o-W_\infty=W_o=1.44\times10^{-6}\,(\mathrm{J})$

9.25 （1） $U_o=\dfrac{q}{4\pi\varepsilon_0 l}-\dfrac{q}{4\pi\varepsilon_0 l}=0$ $\qquad U_c=\dfrac{q}{4\pi\varepsilon_0 3l}-\dfrac{q}{4\pi\varepsilon_0 l}=-\dfrac{q}{6\pi\varepsilon_0 l}$

$A_{oc}=q_0(U_o-U_c)=\dfrac{qq_0}{6\pi\varepsilon_0 l}$



（2） $A_{c\infty}=-q_0(U_c-U_\infty)=\dfrac{qq_0}{6\pi\varepsilon_0 l}$

9.28 （a） $E_1=0$ $\qquad (r<R_1)$

$E_2=\dfrac{Q_1}{4\pi\varepsilon_0 r^2}$ $\qquad (R_1<r<R_2)$ $\qquad E_3=\dfrac{Q_1+Q_2}{4\pi\varepsilon_0 r^2}$ $\qquad (r>R_2)$

$U_1=\int_{0.2}^{0.3}\boldsymbol{E}_2\cdot\mathrm{d}\boldsymbol{r}+\int_{0.3}^{\infty}\boldsymbol{E}_3\cdot\mathrm{d}\boldsymbol{r}=900\,(\mathrm{V})$ $\qquad U_2=\int_{0.5}^{\infty}\boldsymbol{E}_3\cdot\mathrm{d}\boldsymbol{r}=450\,(\mathrm{V})$

（b） $U_{小}=\dfrac{Q_1}{4\pi\varepsilon_0 r}$ $\qquad U_{大}=\dfrac{Q_2}{4\pi\varepsilon_0 r}$

$U_1=\dfrac{Q_1}{4\pi\varepsilon_0\times0.2}+\dfrac{Q_2}{4\pi\varepsilon_0\times0.3}=900\,(\mathrm{V})$ $\qquad U_2=\dfrac{Q_1}{4\pi\varepsilon_0\times0.5}+\dfrac{Q_2}{4\pi\varepsilon_0\times0.5}=450\,(\mathrm{V})$

9.29 $\qquad \mathrm{d}U=\dfrac{\mathrm{d}q}{4\pi\varepsilon_0 x}=\dfrac{\lambda\mathrm{d}x}{4\pi\varepsilon_0 x}$

$U_{AB}=U_{CD}=\int_R^{2R}\dfrac{\lambda\mathrm{d}x}{4\pi\varepsilon_0 x}=\dfrac{\lambda}{4\pi\varepsilon_0}\ln2$ $\qquad U_{BC}=\int_0^{\pi}\dfrac{\lambda R\mathrm{d}\theta}{4\pi\varepsilon_0 R}=\dfrac{\lambda}{4\varepsilon_0}$

$U_0=U_{AB}+U_{BC}+U_{CD}=\dfrac{\lambda}{4\pi\varepsilon_0}(2\ln2+\pi)$

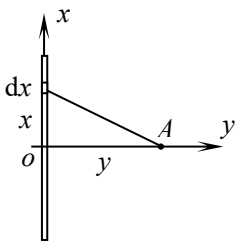
9.30 $\qquad U_+=\dfrac{q}{4\pi\varepsilon_0\sqrt{(x-l/2)^2+R^2}}$ $\qquad U_-=\dfrac{-q}{4\pi\varepsilon_0\sqrt{(x+l/2)^2+R^2}}$

$U=U_++U_-=\dfrac{q}{4\pi\varepsilon_0}(\dfrac{1}{\sqrt{(x-l/2)^2+R^2}}-\dfrac{1}{\sqrt{(x+l/2)^2+R^2}})$

当 $l<<R$ 时，用泰勒级数在 $l=0$ 处展开，略去 l 的二次项，得： $\qquad U=\dfrac{qlx}{4\pi\varepsilon_0(x^2+R^2)^{3/2}}$

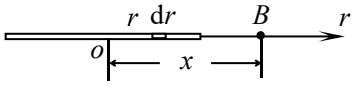
9.31 （1） $U=\int_{-l}^{l}\dfrac{(Q/2l)\mathrm{d}x}{4\pi\varepsilon_0\sqrt{x^2+y^2}}=\dfrac{Q}{4\pi\varepsilon_0 l}\ln\dfrac{\sqrt{l^2+y^2}+l}{y}$

$E=E_y=-\dfrac{\mathrm{d}U}{\mathrm{d}y}=\dfrac{Q}{4\pi\varepsilon_0 y\sqrt{l^2+y^2}}$



（2） $U=\int_{-l}^{l}\dfrac{Q\mathrm{d}r}{8\pi\varepsilon_0 l(x-r)}=\dfrac{Q}{8\pi\varepsilon_0 l}\ln\dfrac{x+l}{x-l}$

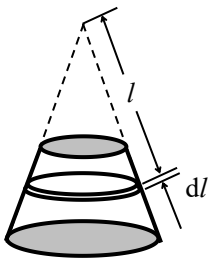
$E=E_x=-\dfrac{\mathrm{d}U}{\mathrm{d}x}=\dfrac{Q}{4\pi\varepsilon_0(x^2-l^2)}$



9.32 （a）锥侧长为变量：　　$\mathrm{d}S = 2\pi r\mathrm{d}l = 2\pi l\sin\theta\mathrm{d}l$

$$\mathrm{d}q = \sigma 2\pi l\sin\theta\mathrm{d}l \qquad \mathrm{d}U = \frac{2\pi l\sigma\sin\theta\mathrm{d}l}{4\pi\varepsilon_0 l}$$

$$U = \int_{R_1/\sin\theta}^{R_2/\sin\theta}\frac{2\pi l\sigma\sin\theta\mathrm{d}l}{4\pi\varepsilon_0 l} = \frac{\sigma}{2\varepsilon_0}(R_2 - R_1)$$



（b）锥高为变量：　　$\mathrm{d}S = 2\pi r\mathrm{d}l = 2\pi x\tan\theta\dfrac{\mathrm{d}x}{\cos\theta}$

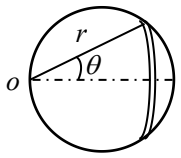
$$U = \int_{R_1/\tan\theta}^{R_2/\tan\theta}\frac{2\pi x\sigma\tan\theta\dfrac{\mathrm{d}x}{\cos\theta}}{4\pi\varepsilon_0\dfrac{x}{\cos\theta}} = \frac{\sigma}{2\varepsilon_0}(R_2 - R_1)$$

（c）半径为变量：　　$\mathrm{d}S = 2\pi r\mathrm{d}l = 2\pi r\dfrac{\mathrm{d}r}{\sin\theta}$　　$U = \int_{R_1}^{R_2}\dfrac{2\pi r\sigma\dfrac{\mathrm{d}r}{\sin\theta}}{4\pi\varepsilon_0\dfrac{r}{\sin\theta}} = \dfrac{\sigma}{2\varepsilon_0}(R_2 - R_1)$

9.33　　$\sigma = \dfrac{Q}{\pi(b^2 - a^2)}$　　$\mathrm{d}q = \sigma\cdot 2\pi r\mathrm{d}r$　　$\mathrm{d}U = \dfrac{\mathrm{d}q}{4\pi\varepsilon_0 r} = \dfrac{\sigma}{2\varepsilon_0}\mathrm{d}r$

$$U = \int_a^b\frac{\sigma}{2\varepsilon_0}\mathrm{d}r = \frac{\sigma}{2\varepsilon_0}(b-a) = \frac{Q}{2\pi\varepsilon_0(a+b)} = 9\times10^3\,(\mathrm{V})$$

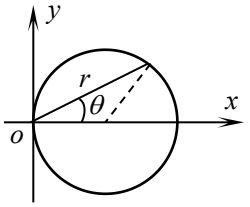
9.34 （a）　$\mathrm{d}q = \sigma\cdot 2\theta r\mathrm{d}r$　　$\mathrm{d}U = \dfrac{\mathrm{d}q}{4\pi\varepsilon_0 r} = \dfrac{\sigma\theta}{2\pi\varepsilon_0}\mathrm{d}r$



$r = 2R\cos\theta$　　$\mathrm{d}r = -2R\sin\theta\mathrm{d}\theta$

$$U = -(\frac{\sigma R}{\pi\varepsilon_0})\int_{\pi/2}^{0}\theta\sin\theta\mathrm{d}\theta = \frac{\sigma R}{\pi\varepsilon_0}$$

（b）　$\mathrm{d}q = \sigma\cdot r\mathrm{d}r\mathrm{d}\theta$　　$\mathrm{d}U = \dfrac{\mathrm{d}q}{4\pi\varepsilon_0 r} = \dfrac{\sigma}{4\pi\varepsilon_0}\mathrm{d}r\mathrm{d}\theta$

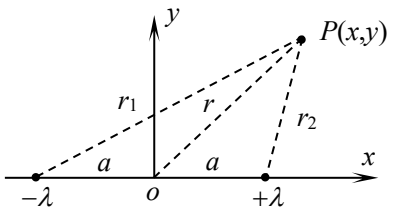


$$U = \frac{\sigma}{4\pi\varepsilon_0}\int_{-\pi/2}^{\pi/2}(\int_0^{2R\cos\theta}\mathrm{d}r)\mathrm{d}\theta = \frac{\sigma}{4\pi\varepsilon_0}\int_{-\pi/2}^{\pi/2}2R\cos\theta\mathrm{d}\theta = \frac{\sigma R}{\pi\varepsilon_0}$$

9.35　　$U_+ = \int_{r_2}^{a}\boldsymbol{E}_+\cdot\mathrm{d}\boldsymbol{r} = \int_{r_2}^{a}\dfrac{\lambda}{2\pi\varepsilon_0 r}\mathrm{d}r = \dfrac{\lambda}{2\pi\varepsilon_0}\ln\dfrac{a}{r_2}$

$$U_- = \int_{r_1}^{a}\boldsymbol{E}_-\cdot\mathrm{d}\boldsymbol{r} = \int_{r_1}^{a}\frac{-\lambda}{2\pi\varepsilon_0 r}\mathrm{d}r = \frac{-\lambda}{2\pi\varepsilon_0}\ln\frac{a}{r_1}$$

$$U_P = U_+ + U_- = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_1}{r_2} = \frac{\lambda}{4\pi\varepsilon_0}\ln\frac{(x+a)^2+y^2}{(x-a)^2+y^2}$$



9.36　　$U_P = \dfrac{q_1}{4\pi\varepsilon_0 r_1} - \dfrac{q_2}{4\pi\varepsilon_0 r_2} = 0$　　$\dfrac{r_1}{r_2} = \dfrac{q_1}{q_2}$

设 A 点为直角坐标系的原点，P 点的坐标为（x、y、z）：

$$\frac{\sqrt{x^2+y^2+z^2}}{\sqrt{(x-d)^2+y^2+z^2}} = \frac{q_1}{q_2}$$

$$x^2 - \frac{2q_1^2 d}{q_1^2 - q_2^2}x + y^2 + z^2 = -\frac{q_1^2 d^2}{q_1^2 - q_2^2}$$

$$(x - \frac{q_1^2 d}{q_1^2 - q_2^2})^2 + y^2 + z^2 = \frac{q_1^2 q_2^2 d^2}{(q_1^2 - q_2^2)^2}$$

